A constant interval of genuine logistic entropy

Problem 00000003363 – lizaixi01

Exact scope. We use the canonical real logistic family $q_a(x) = ax(1-x)$, the whole phase interval $I = [0,1]$, the full usual parameter domain $A = [0,4]$, and topological entropy. The source does not specify its family, domain, entropy kind, or dimension correction. Under these explicit conventions, we refute only its printed "nowhere monotone" conjunct, with monotone having its standard weak, non-strict meaning. We do not claim global Milnor monotonicity, continuity of entropy, a fractal dimension/correction, or a result on parameters restricted to chaos.

Theorem 1. *For every $a \in [1/4, 1/2]$, the actual continuous selfmap $q_a : I \to I$ has $h_{\mathrm{top}}(q_a) = 0$. Therefore the actual entropy function on A is both nondecreasing and nonincreasing on the nonempty open parameter interval $(1/4, 1/2)$. In particular it is not nowhere monotone.*

The actual quadratic family. If $x \in I$, then $0 \le x(1-x) \le 1/4$; hence for every $a \in A$, $0 \le q_a(x) \le 1$. Thus the phase space is the whole usual invariant interval, not an attracting sub-basin. Polynomial operations prove continuity. The formal polynomial is $-aX^2 + aX$, its evaluation is exactly q_a, and its native degree is two whenever $a > 0$. Native real derivatives give

$$q_a'(x) = a(1-2x), \qquad q_a''(x) = -2a.$$

Every local parameter is positive, so $x = 1/2$ is its unique critical point and is nondegenerate. Moreover,

$$q_a(1/2) = a/4.$$

Evaluation at this true phase point proves that the map-valued family is injective in a. The constant entropy interval is not obtained from a constant or degenerate family.

A genuine local contraction. For $x, y \in I$ we have $|1 - x - y| \le 1$, and

$$q_a(x) - q_a(y) = a(x-y)(1-x-y).$$

Consequently, for every $a \in [1/4, 1/2]$,

$$|q_a(x) - q_a(y)| \le \tfrac{1}{2}|x-y|.$$

This proves the native `LipschitzWith (1/2)` property of the actual subtype-valued map. In particular its iterates satisfy $d(q_a^k x, q_a^k y) \le d(x,y)$ for every $k \ge 0$.

Interpretation and primary. Van Strien's author survey, dated 6 July 2012, pp.1–2, discusses ordinary topological entropy and explicitly gives this logistic family. It supplies the standard model and terminology only. Its global results are not Lean axioms, and the argument here uses no external zero-entropy theorem.

All scales and the native entropy limit. We prove a general lemma for a nonempty compact pseudometric space X and an actual nonexpanding selfmap T. Let U be any entourage. Choose $\varepsilon > 0$ with $\{d(x, y) < \varepsilon\} \subseteq U$, and put $V = \{d(x, y) < \varepsilon/2\}$. Then V is symmetric and $V \circ V \subseteq U$. Compactness gives a finite set S whose $\varepsilon/2$-balls cover X. Nonemptiness proves $M = |S| \geq 1$, so its logarithm is finite.

For every point x, choose $y \in S$ with $d(x, y) < \varepsilon/2$. For every iterate k,

$$d(T^k x, T^k y) \leq d(x, y) < \varepsilon/2.$$

Thus this same actual finite set is a dynamical (V, n)-cover for *every* orbit length n. The native library cover inequality and antitonicity in the entourage give, for each integer $n \geq 0$,

$$h_U(T) \;\leq\; h_{V \circ V}(T) \;\leq\; \frac{\log M}{n+1}.$$

The right-hand side tends to zero. The proof checks the native `ENNReal.log` to real logarithm conversion, positive finite cardinal, `EReal` quotient and limit, and native nonnegativity. Taking the supremum over *all* entourages proves the genuine library quantity

$$\texttt{Dynamics.coverEntropy}\ T\ \texttt{Set.univ} = 0.$$

Applying this lemma to each local logistic map proves the theorem; no entropy value is assigned or inferred from finitely sampled iterates.

The open interval contradiction. The endpoints $1/4 < 1/2$ are strictly inside A; their open interval is open in the parameter space and contains $3/8$. Actual entropy is zero at every point of this interval. Hence for every ordered pair of parameters in it, both weak entropy inequalities hold. The formal `NowhereMonotone` predicate excludes such a nonempty open interval in either direction; `printed_nowhere_monotone_false` negates it. This is only the disclosed canonical reading of that conjunct. Other unspecified source clauses and other possible entropy spectra are not settled.

Formal reproduction and evidence. The complete Lean project pins Lean 4.33.0 and Mathlib commit `db584cd6d46c92f209a44c0f1c829460d327499d`. The exact bilingual source is included separately. From `lean/`, run `lean --version`; then `bash verify.sh` runs the build, warning-as-error exhaustive ordinary axiom audit, and official fresh kernel checker. Every owned declaration, including private/generated names, is enumerated and checked. Only `propext`, `Classical.choice`, and `Quot.sound` are allowed; there are no placeholders or extra axioms. Source-bound original logs, execution/affinity data and hashes accompany the submission. Production timings are diagnostic, not controlled efficiency evidence.

Reference. Sebastian van Strien, *Milnor's Conjecture on Monotonicity of Topological Entropy: results and questions*, author PDF, 6 July 2012, pp.1–2.

`https://www.ma.imperial.ac.uk/~svanstri/Files/milnor80-bis.pdf`.